

Fractional heat generator domains on the circle are algebras

A partial answer to Remark 2.9 of arXiv:1901.07477

Automated mathematics run `fa_banach_001`, lane 7

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Status. Candidate partial result; current verdict: *likely valid*. Human review is recommended. For every $0 < \alpha \leq 2$, the packet settles the source's generator-core algebra question affirmatively for the wrapped fractional heat convolution semigroup on the classical compact group $\mathbb{T}$. In fact, the entire generator domain in $C(\mathbb{T})$ is an algebra. The general locally compact quantum-group question remains open.

1 Source question

Skalski and Viselter, *Generating functionals for locally compact quantum groups*, arXiv:1901.07477, Remark 2.9 on page 9, associate to a symmetric convolution semigroup of states a completely positive contraction semigroup $(T_t)_{t \geq 0}$ on the universal quantum-group C^* -algebra, with generator L . Their twisted Fourier cone is denoted $\mathcal{D}_+$. They prove that

$$\text{span}(\mathcal{D}_+) \cap D(L)$$

is a dense selfadjoint invariant core for L , and then state: “We do not know however if it is an algebra” [1, p. 9, Remark 2.9].

We answer this question for a natural one-parameter family on the circle. Identify $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. For $0 < \alpha \leq 2$, let $(\mu_t^{(\alpha)})_{t \geq 0}$ be the symmetric convolution semigroup determined by

$$\widehat{\mu_t^{(\alpha)}}(n) = e^{-t|n|^\alpha}, \quad n \in \mathbb{Z}.$$

Its convolution operators on $C(\mathbb{T})$ are denoted $T_t^{(\alpha)}$, and the positive generator is

$$A_\alpha = -L_\alpha = |D|^\alpha, \quad \widehat{A_\alpha f}(n) = |n|^\alpha \widehat{f}(n).$$

2 Idea of the proof

If $f \in D(A_\alpha)$ and $h = A_\alpha f$, then, modulo its constant Fourier coefficient, f is the periodic Riesz potential $K_\alpha * h$, where $\widehat{K_\alpha}(n) = |n|^{-\alpha}$ for $n \neq 0$. Translation estimates for this kernel give exactly the modulus of continuity

$$\omega_f(r) \lesssim \|h\|_\infty \begin{cases} r^\alpha, & 0 < \alpha < 1, \\ r \log(e/r), & \alpha = 1, \\ r, & 1 < \alpha < 2. \end{cases} \quad (1)$$

For $0 < \alpha < 2$, the product rule for the fractional Laplacian contains the carré-du-champ term

$$\Gamma_\alpha(f, g)(x) = c_\alpha \int_{-\pi}^{\pi} (f(x) - f(x+y))(g(x) - g(x+y))q_\alpha(y) dy,$$

where $q_\alpha(y) \asymp |y|^{-1-\alpha}$ near zero. The product of the two moduli in (1) makes this integral absolutely and uniformly convergent: the local majorant is respectively $r^{\alpha-1}$, $\log^2(e/r)$, or $r^{1-\alpha}$. All are integrable in the stated ranges. Thus $\Gamma_\alpha(f, g)$ is continuous. Poisson smoothing and closedness of A_α then upgrade the formal product rule to the full generator domain.

3 Main result

Theorem 1 (Domain-algebra theorem). *For every $0 < \alpha \leq 2$, the domain $D(A_\alpha)$ of the generator of $(T_t^{(\alpha)})_{t \geq 0}$ on $C(\mathbb{T})$ is closed under pointwise multiplication. More precisely, if $0 < \alpha < 2$ and $f, g \in D(A_\alpha)$, then*

$$A_\alpha(fg) = fA_\alpha g + gA_\alpha f - \Gamma_\alpha(f, g), \quad (2)$$

where $\Gamma_\alpha(f, g) \in C(\mathbb{T})$ is the absolutely convergent integral above.

Corollary 1 (Partial answer to the source). *For the classical locally compact quantum group $\mathbb{T}$ and every wrapped fractional heat semigroup above,*

$$\text{span}(\mathcal{D}_+) \cap D(L_\alpha)$$

is an algebra.

4 Proof

We normalize convolution using Haar probability measure. Constants below may depend on α but not on the functions.

Lemma 1 (Periodic potential modulus). *Let $0 < \alpha < 2$, and let K_α be the periodic distribution with Fourier coefficients*

$$\widehat{K}_\alpha(0) = 0, \quad \widehat{K}_\alpha(n) = |n|^{-\alpha} \quad (n \neq 0).$$

Then $K_\alpha \in L^1(\mathbb{T})$ and, for $0 < |y| \leq 1/2$,

$$\|K_\alpha(\cdot + y) - K_\alpha\|_1 \leq C_\alpha \rho_\alpha(|y|), \quad \rho_\alpha(r) = \begin{cases} r^\alpha, & 0 < \alpha < 1, \\ r \log(e/r), & \alpha = 1, \\ r, & 1 < \alpha < 2. \end{cases} \quad (3)$$

Consequently, if $f \in D(A_\alpha)$, then

$$\omega_f(r) := \sup_{|y| \leq r} \|f(\cdot + y) - f\|_\infty \leq C_\alpha \|A_\alpha f\|_\infty \rho_\alpha(r). \quad (4)$$

Proof. The standard periodic Riesz-kernel estimates, obtained for example by Abel regularization and summation by parts, are

$$\begin{aligned} |K_\alpha(x)| &\leq C|x|^{\alpha-1}, \quad |K'_\alpha(x)| \leq C|x|^{\alpha-2}, & 0 < \alpha < 1, \\ K_1(x) &= -2 \log(2|\sin(x/2)|), \quad |K'_1(x)| \leq C|x|^{-1}, & \alpha = 1, \\ K'_\alpha &\in L^1(\mathbb{T}), & 1 < \alpha < 2. \end{aligned} \quad (5)$$

For $\alpha < 1$, split the translation integral into $\{|x| \leq 2|y|\}$ and its complement. The first part is $O(|y|^\alpha)$ by the bound on K_α ; the mean-value theorem and the derivative bound give the same estimate on the complement. The logarithmic line of (5) gives $O(|y| \log(e/|y|))$ in exactly the same split. For $\alpha > 1$, the $W^{1,1}$ translation estimate gives $O(|y|)$. This proves (3) and also $K_\alpha \in L^1$.

Now put $h = A_\alpha f$. Fourier coefficients, first in the distributional sense and then in $C(\mathbb{T})$, give

$$f - \widehat{f}(0) = K_\alpha * h.$$

Taking a translation difference and using Young's inequality together with (3) proves (4). $\square$

For $0 < \alpha < 2$, let

$$q_\alpha(y) = \sum_{k \in \mathbb{Z}} |y + 2\pi k|^{-1-\alpha}, \quad -\pi < y < \pi, y \neq 0.$$

Up to a positive normalization constant, the usual periodic singular-integral formula reads

$$A_\alpha u(x) = c_\alpha \text{pv} \int_{-\pi}^{\pi} (u(x) - u(x+y)) q_\alpha(y) dy \quad (6)$$

for smooth u ; it is the periodization of the fractional-Laplacian formula [2]. Notice that $q_\alpha(y) \leq C_\alpha |y|^{-1-\alpha}$ near zero and is bounded on compact subintervals avoiding zero.

Let $f, g \in D(A_\alpha)$. Lemma 1 gives

$$\begin{aligned} & |(f(x) - f(x+y))(g(x) - g(x+y))q_\alpha(y)| \\ & \leq C \|A_\alpha f\|_\infty \|A_\alpha g\|_\infty \begin{cases} |y|^{\alpha-1}, & 0 < \alpha < 1, \\ \log^2(e/|y|), & \alpha = 1, \\ |y|^{1-\alpha}, & 1 < \alpha < 2. \end{cases} \end{aligned} \quad (7)$$

Each right-hand side is integrable at zero. The integral defining $\Gamma_\alpha(f, g)$ is therefore absolutely and uniformly convergent. Dominated convergence shows that it is continuous. The same estimate also makes Γ_α continuous as a bilinear map when each domain is given the graph norm $\|u\|_\infty + \|A_\alpha u\|_\infty$.

For $s > 0$, set $f_s = T_s^{(\alpha)} f$ and $g_s = T_s^{(\alpha)} g$. These functions are smooth, and semigroup theory gives

$$f_s \rightarrow f, \quad A_\alpha f_s = T_s^{(\alpha)} A_\alpha f \rightarrow A_\alpha f \quad \text{uniformly,}$$

with the analogous statements for g . Applying (6) to the smooth products gives

$$A_\alpha(f_s g_s) = f_s A_\alpha g_s + g_s A_\alpha f_s - \Gamma_\alpha(f_s, g_s).$$

The right side converges uniformly, by the preceding graph-norm continuity, and $f_s g_s \rightarrow fg$ uniformly. Since A_α is closed, $fg \in D(A_\alpha)$ and (2) holds. This proves Theorem 1 for $0 < \alpha < 2$.

For $\alpha = 2$, the generator domain on $C(\mathbb{T})$ is $D(A_2) = C^2(\mathbb{T})$: distributionally $A_2 f = -f''$, so continuity of $A_2 f$ allows two integrations and yields a classical second derivative. This space is plainly an algebra. The theorem follows.

For Corollary 1, the scaling group of the classical compact group $\mathbb{T}$ is trivial, and the source identifies $\mathcal{D}_+ = A(\mathbb{T}) \cap P(\mathbb{T})$, where $A(\mathbb{T})$ is the Wiener algebra and $P(\mathbb{T})$ the positive-definite cone. Every absolutely summable Fourier sequence is a linear combination of four nonnegative sequences, so $\text{span}(\mathcal{D}_+) = A(\mathbb{T})$. Theorem 1 makes the larger space $D(L_\alpha) = D(A_\alpha)$ an algebra, hence its intersection with $A(\mathbb{T})$ is an algebra as well.

5 Verification and limitations

Proof audit. The proof uses three checkable inputs: the L^1 translation modulus of the periodic Riesz kernel, the positive periodized singular-integral kernel, and closedness of the semigroup generator. The exponents in (7) were checked separately on the three ranges divided by $\alpha = 1$. No computational experiment is used as proof evidence.

Human-review focus. The recommended review points are the endpoint estimate $\rho_1(r) = r \log(e/r)$, graph-norm continuity of Γ_α , and the identification $\text{span}(\mathcal{D}_+) = A(\mathbb{T})$ in the source's conventions.

Scope. This does not resolve the source's question for an arbitrary convolution semigroup, even on $\mathbb{T}$, and does not resolve it for general locally compact quantum groups. The proof succeeds because the stable symbol gives sharp potential-kernel regularity matching the singularity of its carré du champ.

6 Novelty check

A bounded search checked the exact sentence from Remark 2.9, the source title together with generator-domain and algebra keywords, and fractional Laplacian domain-algebra terminology. It located broad Sobolev algebra and fractional Schauder results, including [2], but no later paper stating the source identification or Theorem 1 on $C(\mathbb{T})$. The regularity ingredients are standard, and the theorem may be an unstated consequence of known Schauder theory. Novelty is therefore *provisional*, not established.

References

- [1] A. Skalski and A. Viselter, *Generating functionals for locally compact quantum groups*, International Mathematics Research Notices 2021 (2021), 132–159; arXiv:1901.07477.
- [2] P. R. Stinga, *User's guide to the fractional Laplacian and the method of semigroups*, in *Handbook of Fractional Calculus with Applications*, Vol. 2, De Gruyter, 2019, 235–266; arXiv:1808.05159.